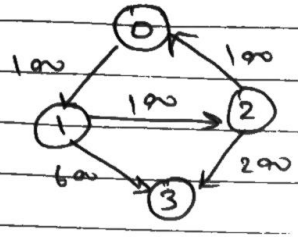
Name : Amol Saurav
UID : 23BCS10067
Section: KRG-2B

Experiment-7

8) Find minimum number of flights with at most K steps



The optimal path with at most 1 stop from city 0 to 3 has cost 700.

Approach:

1. Initialize distance array
   dist[i] = minimum cost to reach node i
   Set all = ∞, except dist[src] = 0

2) Create queue
   Store : {node, cost, stops}
   push : {src, 0, 0}

3) Start BFS traversal

4) Explore neighbors
   for each neighbor:
   Calculate new cost
   if new cost < dist[neighbor]
   → update distance
   → push into queue with stops + 1

5) Return answer
   if dist[dst] == ∞ → return -1
   Else → return dist[dst]

Code :-

```
int findCheapestPrice(int n, vector<vector<int>>& flights, int src,
                      int dst, int k) {
    vector<vector<pair<int, int>>> adj(n);
```

```
for(auto & f: flights){
        adj[f[0]].push_back({f[1], f[2]});
}
vector<int> dist(n, INT_MAX);
dist[src]=0;

queue<tuple<int, int, int>> q;
q.push({src, 0, 0});

while(!q.empty()){
    auto [node, cost, stops]= q.front();
    q.pop();
    if(stops>k) continue;

    for(auto & it: adj[node]){
        int nextNode = it.first;
        int price = it.second;
        if(cost + price < dist[nextNode]){
            dist[nextNode] = cost + price;
            q.push({nextNode, dist[nextNode],
                    stops+1});
        }
    }
}
return dist[dst]==INT_MAX ? -1 : dist[dst];
}
```

Time Complexity : $O(E*K)$
Space Complexity: $O(V+E)$